



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2024-25



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 05

Aim: Implementation of K-mean Clustering.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook

Theory:

K-Means Clustering

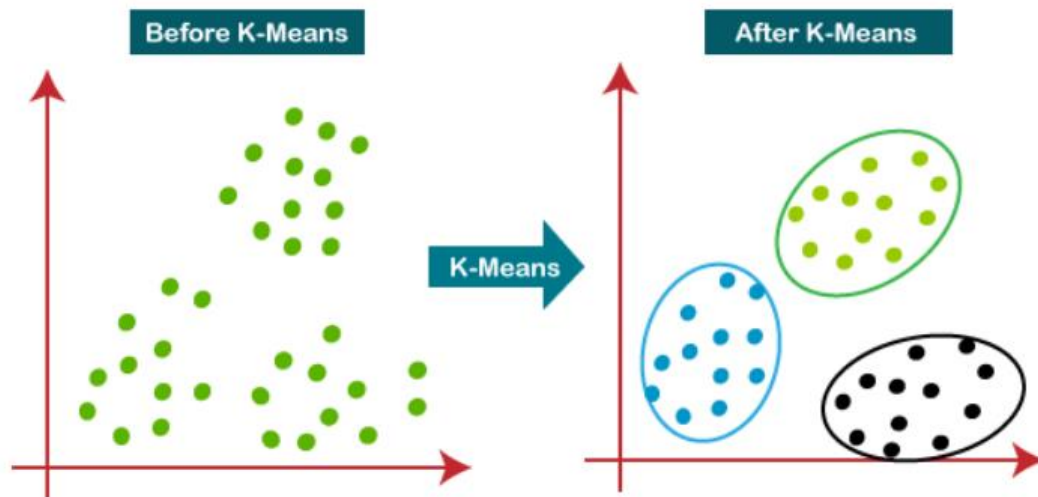
- K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, as if $K=2$, there will be two clusters, and for $K=3$, there will be three clusters, and so on.
- It is an iterative algorithm that divides the unlabeled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.
- It allows us to cluster the data into different groups and a convenient way to discover the categories of groups in the unlabeled dataset on its own without the need for any training.
- It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.
- The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters.
- The value of k should be predetermined in this algorithm.

The k-means clustering algorithm mainly performs two tasks:

- 1) Determines the best value for K center points or centroids by an iterative process.
- 2) Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.

Hence each cluster has datapoints with some commonalities, and it is away from other clusters.

The below diagram explains the working of the K-means Clustering Algorithm:



The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

Program:

```
from sklearn.cluster import KMeans
import numpy as np
import matplotlib.pyplot as plt

# Generate some sample data
X = np.array([[1, 2], [1, 4], [1, 0], [4, 2], [4, 4], [4, 0]])

# Number of clusters
k = 2

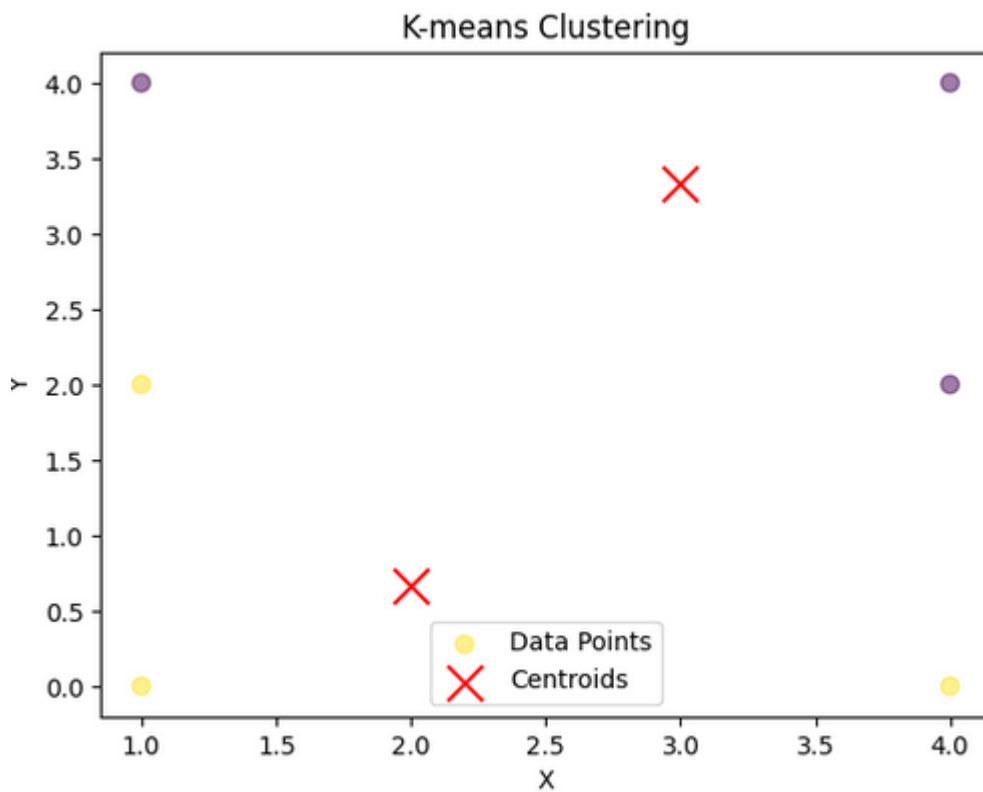
# Create KMeans instance
kmeans = KMeans(n_clusters=k)

# Fit the model to the data
kmeans.fit(X)

# Get cluster centroids and labels
centroids = kmeans.cluster_centers_
labels = kmeans.labels_
```

```
# Plot the data points and centroids
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='viridis', s=50, alpha=0.5, label='Data Points')
plt.scatter(centroids[:, 0], centroids[:, 1], c='red', marker='x', s=200, label='Centroids')
plt.title('K-means Clustering')
plt.xlabel('X')
plt.ylabel('Y')
plt.legend()
plt.show()
```

Output:



Conclusion: The implementation of K-Means Clustering demonstrates how unsupervised learning can effectively group similar data points into clusters based on their features.